UC DAVIS ANESTHESIOLOGY • NAPA CME 2025

Beyond the Hype:

Safe, Supervised AI Use in Anesthesia

"AI reasons. Humans judge. Verify. Decide. Document."

CORE PRINCIPLE:

Verify.
Decide.
Document.

AI reasons. You judge. Non-delegable clinical responsibility.

RESOURCES INCLUDED IN THIS GUIDE

1. **Clinical AI Decision Card**
   Verify → Decide → Document · Bedside Quick Reference
2. **Department Governance Readiness**
   Pre-Deployment Gap Analysis · 25-Item Checklist
3. **PHI Environment Decision Flowchart**
   Is This AI Use In-Bounds? · Step-by-Step Guide
4. **90-Day Pilot Roadmap Template**
   From Problem Definition to Governed Deployment
5. **How to Build a Department AI Guide**
   8-Component Framework · UC Davis Task Force as a Model
6. **Calibration vs. Discrimination**
   Understanding What AI Performance Metrics Actually Tell You

PRESENTER

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RESOURCE 1

Clinical AI Decision Card

Verify → Decide → Document · Bedside Quick Reference

① VERIFY	② DECIDE	③ DOCUMENT
Check the Data	**Make the Call**	**Record the Plan**
Does the signal match independent clinical assessment? **Ask yourself:** • Is the signal plausible? • Could it be artifact? • What does the patient look like? • Can I trace the source?	*AI informs. You own the judgment boundary.* **Consider:** • Patient context + history • Competing harms • Reversibility of action • Surgeon/team communication	*Chart your reasoning, not the algorithm.* **Write:** • Alert evaluated (name it) • My independent assessment • My decision + rationale • *Not: "AI recommended; agreed"*

DOCUMENTATION TEMPLATE — COPY INTO YOUR NOTE

TEMPLATE

"[Tool name] alert evaluated. [Clinical finding]. [Relevant vitals/context]. [Conclusion: artifact suspected / clinically consistent / etc.]. [Action or no action and reason]."

CONCRETE EXAMPLE

"HPI alert 87 evaluated. A-line waveform damped. MAP 68, stable. Artifact suspected. No vasopressor at this time. Will reassess in 5 minutes."

COGNITIVE AI VS. CLINICAL AI — KNOW WHAT YOU'RE USING

COGNITIVE AI

Summarizes · Explains · Drafts · Compares

- ChatGPT, Claude, Gemini (consumer)
- Use to think and brainstorm
- Never for patient-specific decisions
- No PHI outside approved pathways

Rule: Use it to think. Never use it to decide.

CLINICAL AI

Detects · Predicts · Overlays · Flags

- HPI, echo overlay, ambient note
- FDA-cleared, enterprise-deployed
- Requires VDD supervision every use
- Prediction is not permission to act

Rule: AI informs. You judge. Always document.

AI reasons. Humans judge. Verify. Decide. Document.

RESOURCE 2

Department Governance Readiness Checklist

Pre-Deployment Gap Analysis · SEDASYS Failure Prevention Framework

Use this checklist before any AI tool moves from pilot to standard deployment. An unanswered item is a governance gap — not a bureaucratic hurdle.

A. CLINICAL OWNERSHIP — WHO ANSWERS WHEN IT FAILS?

- ☐ **A named physician is designated as Clinical Owner for this tool**
  Name & title documented in writing
- ☐ The Clinical Owner has reviewed the tool's validation study and understands its failure modes
- ☐ The Clinical Owner has authority to suspend the tool without committee approval
- ☐ A backup Clinical Owner is named (coverage during leave/absence)
- ☐ The Clinical Owner role is reviewed and reconfirmed annually

B. KILL SWITCH & ESCALATION — WHAT HAPPENS WHEN SOMETHING GOES WRONG?

- ☐ **A suspension protocol exists: who initiates, how, and in what timeframe**
- ☐ The kill switch authority is documented and known to frontline staff
- ☐ An escalation path is defined for safety events: clinical owner → department chief → risk management
- ☐ A post-suspension review process is defined (what triggers reinstatement?)
- ☐ The suspension protocol has been tested in a tabletop exercise or simulation

C. LOCAL VALIDATION — WAS IT TESTED ON YOUR PATIENTS?

- ☐ **The tool was validated in a local pilot before full deployment**
- ☐ Pilot cohort reflected our actual case mix and patient population
- ☐ Performance metrics were measured in our workflow, not only the vendor's dataset
- ☐ Alert threshold was calibrated to local false-positive/false-negative tolerance
- ☐ Local validation results are documented and accessible for review

D. DOCUMENTATION STANDARD — IS THE VDD PROTOCOL IN PLACE?

- ☐ **A documentation template exists for this specific tool (one or two sentences)**
- ☐ Template has been shared with all clinical staff who use this tool

- ☐ The template captures: signal acknowledged, independent assessment, decision rationale
- ☐ The prohibited phrase "AI recommended X; agreed" is explicitly flagged in onboarding
- ☐ New staff receive tool-specific documentation orientation before first use

E. ONGOING MONITORING — WHO CHECKS AFTER GO-LIVE?

- ☐ **A metric set was established before go-live (override rate, alert burden, adverse events)**
- ☐ A baseline measurement was recorded at launch
- ☐ A review cadence is scheduled (minimum: quarterly)
- ☐ A Technical Owner is responsible for model updates and performance monitoring
- ☐ Drift detection: process exists for flagging when alert rate or accuracy changes unexpectedly

READINESS VERDICT	**All items checked:** Approved for deployment **1–3 items unchecked:** Conditional — document gap and assign owner **4+ items unchecked:** Not ready — governance work required first

Reminder: *FDA clearance permits selling — not deploying unsupervised. Governance prevents deployment disasters.*

RESOURCE 3

PHI Environment Decision Flowchart

Is This AI Use In-Bounds? · Step-by-Step Decision Guide

Work through each step from top to bottom. Stop at the first red outcome. The guiding principle: **environment matters more than branding.**

1

Are you about to enter patient-identifiable information (name, MRN, DOB, diagnosis, medication, or any PHI)?

YES → PHI rules do not apply. Proceed to Step 2.

NO → Continue to Step 2 immediately.

2

Is this tool enterprise-approved by your institution?

Enterprise-approved means: procured through IT/legal/compliance, with a signed Business Associate Agreement (BAA) if PHI is involved.

YES → Continue to Step 3.

NO → STOP. Do not enter PHI. Consumer tool — brainstorm and outline only.

3

Is there a signed Business Associate Agreement (BAA) with the vendor?

A BAA is the legal agreement under HIPAA that governs how the vendor handles PHI. Without it, PHI cannot lawfully flow to the tool.

YES → Continue to Step 4.

NO → STOP. Contact your compliance or privacy officer before proceeding.

4

Is this a clinical AI tool (predictive/monitoring/diagnostic) or a cognitive AI tool (drafting/summarizing)?

YES → COGNITIVE: Permitted for thinking and drafting. Apply Verify-Decide-Document before acting. Independent review required.

NO → CLINICAL: Continue to Step 5.

5

Has this clinical AI tool passed the Governance Readiness Checklist (Resource 2) for your department?

YES → CLEARED. Proceed with Verify → Decide → Document every use.

NO → STOP. Complete governance checklist. Report gap to department AI owner.

STOP — RED	CONDITIONAL — YELLOW	CLEARED — GREEN
No enterprise approval, no BAA, or governance incomplete. Do not enter PHI. Escalate.	Cognitive AI with enterprise approval. No PHI for clinical decisions. Brainstorm and draft only.	Enterprise-approved, BAA signed, governance complete. Apply VDD every use.

Key principle: *Environment matters more than branding. The name of the tool does not determine compliance — the deployment context does.*

RESOURCE 4

90-Day Pilot Roadmap Template

From Problem Definition to Governed Deployment · Department Leader Planning Tool

Fill in the fields below before your first department meeting on AI deployment. You do not need to solve institutional AI governance before Monday. You need **one tool, one protocol, one owner.**

TOOL BEING PILOTED

_____

TOOL TYPE

☐ Prediction ☐ Overlay ☐ Ambient Note ☐ Evidence

PHASE 1 — DAYS 1–30: DEFINE & VET

DEFINE THE PROBLEM STATEMENT

Not 'we should adopt AI' — a specific, measurable clinical gap.

The problem this tool addresses:

_____

_____

_____

FIVE PRE-DEPLOYMENT QUESTIONS

☐ Does it represent my patients?

☐ Who owns failure?

☐ Can I defend follow / override?

☐ Was it validated here?

☐ Who owns the kill switch?

Unanswered items: _____ / 5

PHASE 2 — DAYS 31–60: PILOT LOCALLY

PILOT PARAMETERS

Cohort size: _____ cases

Setting: _____

Duration: _____ weeks

Pilot Lead: _____

DOCUMENTATION STANDARD

Draft your 2-sentence note template:

_____

_____

_____

FEEDBACK COLLECTION

Ask each pilot user:

- Alert burden (1–10)?
- Time added per case?
- Perceived accuracy?
- Any near-misses?

PHASE 3 — DAYS 61–90: MONITOR, OWN, AND REVIEW

ASSIGN OWNERSHIP BEFORE GO-LIVE

Clinical Owner: _____

Technical Owner: _____

Operations Owner: _____

Kill Switch Authority: _____

Review Cadence: ☐ Monthly ☐ Quarterly ☐ Semi-annual

BASELINE METRICS (RECORD AT GO-LIVE)

Alert rate (per 100 cases): _____

Override rate: _____

Documentation compliance: _____

Clinician burden score: _____

90-day review date: _____

Do not jump from demo to deployment. Vet — Pilot — Monitor — Review.

RESOURCE 5

How to Build a Department AI Guide

Anatomy of an effective AI strategic roadmap · UC Davis Anesthesiology Task Force Report as a working example

What is a department AI guide?

A department-specific AI guide is a written strategic roadmap that answers three questions for your specialty:

1. **What AI tools are relevant to us — and how do we evaluate them?**
2. **How do we deploy and govern them responsibly?**
3. **How do we train our people to use them safely?**

It is not a policy manual, a vendor contract, or a technology wish list. It is a living document that structures the conversation your department needs to have before AI tools are in routine use.

Why does your department need one?

- **AI tools are already arriving:** without department-level guidance — in vendor contracts, Epic updates, and residents' pockets.
- **Governance vacuums cause harm:** SEDASYS failed not from bad technology but from unclear ownership and no escalation path.
- **Clinicians need a framework:** 76% of UC Davis Anesthesiology staff wanted AI training, but 50% had never used AI professionally.
- **External frameworks don't fit:** NIST, AMA, and FDA guidance is general. Your specialty's workflows, risks, and data are specific.

EIGHT COMPONENTS OF AN EFFECTIVE DEPARTMENT AI GUIDE

Each section below identifies what belongs in that component, the key questions it must answer, and what the UC Davis Anesthesiology Task Force put there. Use it as a model — then fill it with your specialty's specifics.

1 Needs Assessment + Stakeholder Survey

WHAT IT CONTAINS

- Survey of current AI use across roles (attendings, residents, trainees, nurses, APPs, admin)
- Quantified adoption gap: personal vs. professional AI use
- Barriers by role: training gaps, legal concerns, reliability fears
- Perceived benefits and highest-priority use cases
- Qualitative themes from open-ended responses

KEY QUESTIONS TO ANSWER

- How many staff are using AI right now — for what tasks?
- Where is the biggest gap between interest and actual use?
- What are the top fears? (accuracy, liability, over-reliance?)
- Which use cases have the most cross-role support?
- What training format does your staff prefer?

UC Davis example: *34-person survey (May 2025): 76% wanted training; 50% had never used AI professionally; top barrier was lack of training (32%)*

2 Task Force Composition + Methodology

WHAT IT CONTAINS

- Named membership list with role and domain (faculty, trainees, nurses, IT, ethics)
- Meeting cadence and duration (e.g., biweekly for 6 months)
- Structured phases: landscape review → use case prioritization → policy → synthesis
- Explicit statement of AI-assisted drafting with human review
- Clear framing: aspirational roadmap, not mandates

KEY QUESTIONS TO ANSWER

- Who has a seat at the table — and who is missing?
- How did you gather evidence? (literature, peer institutions, surveys?)
- What process did you use to prioritize recommendations?
- How will the document be updated over time?
- Who signed off on the final version?

UC Davis example: *10-person task force: faculty, CA3 resident, CRNA, IT security lead, department chair — biweekly meetings Jan–June 2025*

3 Governance & Foundational Framework

WHAT IT CONTAINS

- Approved platform(s) for AI use (HIPAA-compliant, BAA confirmed)
- AI Ethics Oversight Committee: composition, authority, review cadence
- Human-in-the-Loop policy: clinicians remain the final decision-makers
- PHI rules: what can and cannot go into which tools
- Peer institution collaboration plan for shared standards

KEY QUESTIONS TO ANSWER

- Which platforms are approved? Who approved them?
- Who reviews new AI tools before deployment?
- What is the kill switch process — who can suspend a tool?
- How is PHI protected across all tool types?
- What is the escalation path when something goes wrong?

UC Davis example: *Microsoft Azure OpenAI as HIPAA-compliant foundation; AI Ethics Oversight Committee proposed with authority to suspend tools; Human-in-the-Loop policy formalized*

4 Clinical Applications & Pilot Plan

WHAT IT CONTAINS

- Prioritized use cases with status (proposed, in development, active)
- Named lead stakeholders for each pilot
- Sandbox environment for pre-deployment testing
- Success and safety metrics for each use case
- Real-time performance dashboard plan post-deployment

KEY QUESTIONS TO ANSWER

- What are the 3–5 highest-value clinical AI use cases for your specialty?
- Which are ready for piloting now vs. longer-term?
- Who owns each pilot — clinical lead, technical lead, operations lead?
- What does 'success' look like? What triggers suspension?
- How will you capture and respond to clinician feedback?

UC Davis example: *Five pilot use cases: AI documentation (C8 Health), perioperative risk stratification, real-time decision support, PACU discharge tool, ambient scribing*

5 Education & Training Plan

WHAT IT CONTAINS

- Progressive competency map by training level (PGY-1 through PGY-4)
- Integration with existing didactics, PBLDs, and simulation curriculum
- Optional certificate track with modules, simulation, and scholarly project
- AI-enhanced simulation scenarios with real-time tool feedback
- Supervised clinical use protocol with required documentation of reasoning

KEY QUESTIONS TO ANSWER

- What does an AI-competent clinician in your specialty know and do?
- Where can AI content be embedded in existing teaching without adding burden?
- How will you teach critical appraisal, not just tool use?
- What's the minimum viable AI training for every clinician?
- How will you assess competency — observation, reflection, project?

UC Davis example: *Four-year progressive curriculum: ML basics (PGY-1) → CDS tools + ROC curves (PGY-3) → bias/ethics/FDA (PGY-4); ACGME milestone alignment in Appendix H*

6 Research Integration Plan

WHAT IT CONTAINS

- Mentored AI scholarly activity track (aligned with ACGME requirements)
- Longitudinal research elective with hands-on model work
- Multi-institutional collaboration and data-sharing agreements
- Annual AI symposium plan (resident + faculty presentations)
- External funding targets: NIH, DoD, AHRQ

KEY QUESTIONS TO ANSWER

- How will trainees contribute to AI research — not just consume it?
- What institutional data infrastructure exists for model development?
- Which faculty have AI research expertise who can mentor?
- Are there peer institutions you could collaborate with on validation studies?
- What QI and scholarly projects are already underway that AI could accelerate?

UC Davis example: *AI Scholarly Activity Track for residents; longitudinal elective with CeDAR collaboration; annual symposium; NIH/AHRQ funding targets identified*

7 Ethical & Regulatory Framework

WHAT IT CONTAINS

- Institutional ethical AI principles (augmentation, transparency, equity, accountability)
- Relevant national frameworks: NIST AI RMF, AMA policy, FDA GMLP, AAMC competencies
- UC/health system-specific guidance on generative AI
- Survey-derived stakeholder concerns (transparency, liability, over-reliance)
- Commitments: bias audits, explainability standards, ethics education

KEY QUESTIONS TO ANSWER

- What are your institution's official principles for AI use?
- Which national frameworks apply to your specialty?
- What specific ethical concerns did your staff raise?
- How will you audit for algorithmic bias?
- Who is responsible for ethics oversight — and what is their authority?

UC Davis example: *Ethics section cites NIST, AMA, FDA, AAMC; 31% of survey respondents cited ethical/legal concerns as top barrier; AI Ethics Committee charter drafted*

8

Appendices: Tools, Metrics, Prompts, Policies

WHAT IT CONTAINS

- AI pilot use case table (tool, description, status, lead)
- Evaluation metrics across: clinical impact, safety, equity, scalability, ethics
- Prompt library with use cases, sample prompts, and responsible use guidance
- Curriculum integration plan with recommendation-to-action mapping
- Referenced institutional policies and resources with links/citations

KEY QUESTIONS TO ANSWER

- What tools are in use or under evaluation right now?
- What metrics will tell you if a tool is working — or harming?
- What prompts are safe and effective for common use cases?
- What internal policies govern your AI work?
- How will you keep this section current as tools change?

UC Davis example: *10 appendices: use case table, curriculum plan, ethical frameworks, policy index, prompt library, evaluation metrics, ACGME milestone map, survey findings, implementation proposals*

BUILD YOUR OWN: PRE-WRITING CHECKLIST

Before You Start Writing

☐ Identify a task force lead with protected time

☐ Recruit a cross-role group (attendings, trainees, nurses/APPs, IT, ethics/legal)

☐ Run a brief needs assessment survey
10–15 questions; aim for >50% response rate

☐ Catalog what AI tools are already in use in your department

☐ Identify your institution's approved platforms and PHI policies

☐ Review 1–2 peer institution AI guides for benchmarking

☐ Set a realistic timeline
3–6 months for a first draft is common

Before You Publish

☐ Each pillar has named ownership (not just 'the committee')

☐ Every clinical pilot has a kill switch and escalation path defined

☐ The document is explicitly framed as aspirational, not a mandate

☐ AI assistance in drafting is disclosed

☐ Department chair has reviewed and endorsed

☐ A review cadence is built in
Recommend: annual update + triggered by major tool deployment

☐ Staff know where to find it and how to submit feedback

THREE PRINCIPLES FOR ANY DEPARTMENT AI GUIDE

Augmentation, not automation

Every recommendation should reinforce that clinicians remain the final decision-makers. The goal is to enhance judgment — never to bypass it. Say this explicitly, repeatedly, and in plain language.

Governance before deployment

Own the kill switch before you flip the on switch. Regulatory clearance permits selling — not deploying unsupervised. Define who owns failure before a tool goes live, not after something goes wrong.

Specific beats comprehensive

A 5-page guide with one named clinical owner and one documented pilot beats a 40-page framework with no action owner. Start with the smallest version that still answers all eight components — then iterate.

RESOURCE 6

Calibration vs. Discrimination

Understanding What AI Performance Metrics Actually Tell You · Clinician Reference

Vendors present AUROC. Patients need calibration. These are different questions — and confusing them is one of the most common errors in clinical AI evaluation.

DISCRIMINATION

THE QUESTION IT ANSWERS:

"Can the model rank high-risk patients above low-risk ones?"

HOW IT'S REPORTED:

AUROC (Area Under the Receiver Operating Curve), also called AUC or c-statistic. Ranges from 0.5 (coin flip) to 1.0 (perfect).

WHAT A HIGH AUROC TELLS YOU:

The model separates high-risk from low-risk patients in the test dataset. It says nothing about whether the predicted probabilities are accurate.

ANALOGY:

A model correctly ranking a very sick patient above a healthy one has good discrimination — even if it predicts 99% risk when the true risk is 60%.

CALIBRATION

THE QUESTION IT ANSWERS:

"When the model says 80% probability, does the event actually happen 80% of the time — in my patients?"

HOW IT'S REPORTED:

Calibration plots, Hosmer-Lemeshow test, Expected-to-Observed (E/O) ratio. Rarely shown in vendor presentations.

WHAT GOOD CALIBRATION TELLS YOU:

The predicted probabilities are trustworthy as absolute numbers. A 30% alert means something different than a 90% alert.

ANALOGY:

A weather app ranking rainy days above sunny ones but saying 90% rain when actual rate is 40% leaves you carrying an umbrella unnecessarily.

WHY THIS MATTERS CLINICALLY: THE DANGEROUS COMBINATION

A model can have excellent discrimination AND terrible calibration. AUROC 0.87 does not mean the predicted probabilities are accurate in your patients.

Real-world implication:

The Hypotension Prediction Index (HPI) may rank your pre-hypotensive patients correctly above stable patients (good discrimination) but consistently over-predict the severity of risk in your elderly, vasopressor-naïve population (poor local calibration). You'll see a lot of alerts at a severity level that doesn't match clinical observation. That's not a reason to ignore the tool — but it is a reason to demand local calibration data.

READING A CALIBRATION PLOT

WHAT THE AXES MEAN

X-axis: Predicted probability (what the model says)

Y-axis: Observed event rate (what actually happened)

Perfect calibration = the diagonal line

If the model predicts 40% risk for a group and 40% of them have the event: perfect calibration for that decile.

Overconfident model:

Points fall below the diagonal. Model predicts higher probability than actually occurs. More false alarms.

WHAT TO LOOK FOR FROM YOUR VENDOR

- Was a calibration plot published alongside the AUROC?
- Was calibration assessed on an external validation cohort — not just the training data?
- Is the E/O ratio (Expected to Observed) reported? Values near 1.0 are well-calibrated; >1.2 or <0.8 indicate miscalibration.
- Was recalibration performed after deployment in any external site?
- Does the vendor provide tools for local recalibration?

THREE QUESTIONS TO ASK EVERY VENDOR

Q1

"What is the calibration of this model in your external validation cohort, and do you have a calibration plot we can review?"

Red flag: vendor only shows AUROC and cannot produce a calibration plot.

Q2

"Was this model validated on a population similar to ours — in terms of age, ASA class, procedure type, and EMR platform?"

Red flag: validation cohort is academic, predominantly young, or from a single institution with different workflows.

Q3

"When was the model last recalibrated? Do you offer local recalibration, and what does that process look like?"

Red flag: vendor cannot define a recalibration schedule or says it is not needed post-deployment.

High ROC ≠ Local Trustworthiness. The model that ranks well in their data may not predict accurately in yours.

QUICK REFERENCE: DISCRIMINATION VS. CALIBRATION

	DISCRIMINATION	CALIBRATION
Question	Can it rank risk correctly?	Are the probabilities accurate?
Metric	AUROC / AUC / c-statistic	Calibration plot, E/O ratio, H-L test
Good value	0.80–1.0	Points on diagonal; E/O near 1.0
Vendor shows it?	Almost always	Rarely without asking
Local variation	Moderate	High — varies with case mix
Fails silently	Rarely	Frequently after deployment

CALIBRATION PLOT — FIGURE REFERENCE

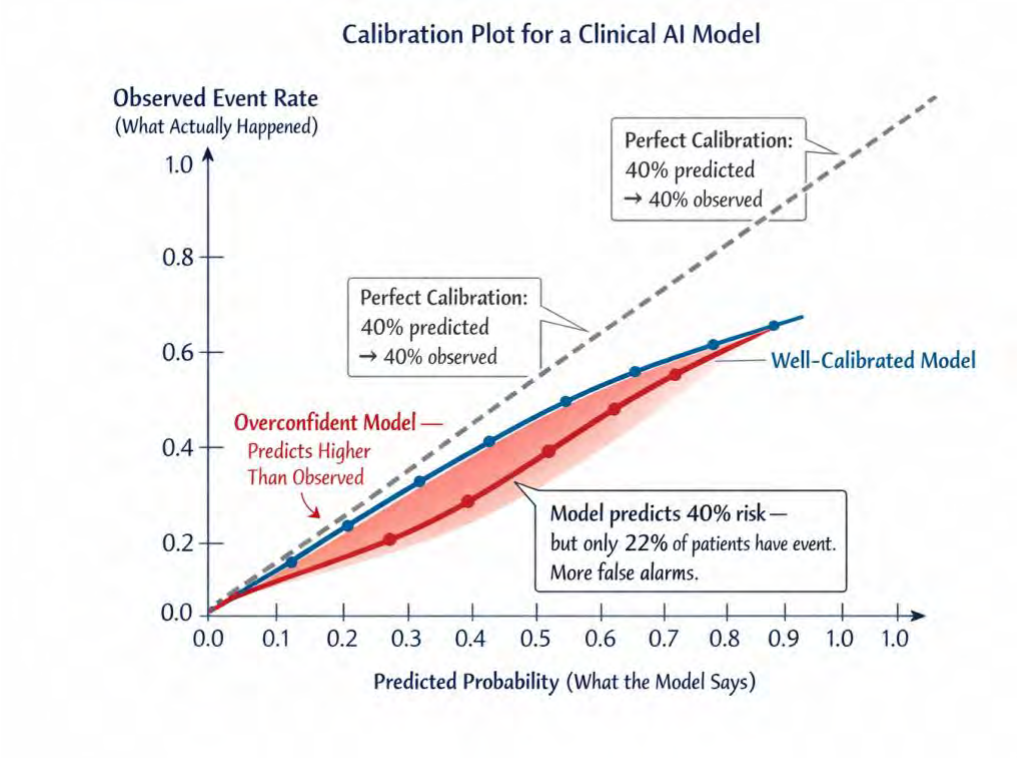

Figure 1. *Calibration plot assessing the reliability of predicted probabilities from a clinical AI model. The dashed diagonal line represents perfect calibration. The blue line shows a well-calibrated model; the red region shows an overconfident model.*

GOING DEEPER: KEY POINTS FOR ADVANCED READERS

Confusing discrimination with calibration A model can demonstrate strong discrimination, as measured by a high area under the receiver operating characteristic curve (AUROC or c-statistic), while exhibiting poor calibration. Discrimination assesses the model's ability to rank patients correctly by their relative risk—that is, whether higher-risk patients receive higher predicted probabilities than lower-risk patients. However, this metric does not ensure that the predicted probabilities accurately reflect the observed event rates in the population. For example, a model with an AUROC of 0.85 may reliably distinguish high-risk from low-risk individuals but systematically overestimate or underestimate absolute risks across probability ranges. In clinical practice, overreliance on discrimination alone can lead to misinterpretations of model performance, particularly when absolute risk estimates inform treatment decisions or patient counseling.

Overemphasizing the calibration slope without the intercept. A calibration slope near 1.0 is frequently interpreted as evidence of good calibration, but this assessment remains incomplete without considering the intercept. The calibration slope reflects the degree to which predicted risks vary appropriately with observed risks (a slope of 1 indicates no shrinkage or expansion of the risk distribution), whereas the intercept quantifies systematic bias in average predicted risk relative to observed events (calibration-in-the-large). A slope of 1 combined with a non-zero intercept can result in consistent overestimation or underestimation of risk across the spectrum. Such systematic bias may not distort relative ranking but can lead to inappropriate clinical actions, such as unnecessary interventions in cases of overestimation or missed opportunities in cases of underestimation. Comprehensive evaluation, therefore, requires reporting and interpreting both metrics together.

Failing to recognize the consequences of overconfidence. When the calibration curve lies below the ideal 45-degree diagonal line, the model exhibits overconfidence, predicting higher risks than are actually observed. This pattern generates excess false positives, which in turn promote unnecessary diagnostic tests, treatments, or interventions. In patient care, these consequences include avoidable harms such as procedural complications, medication adverse effects, healthcare resource overuse, and psychological distress from heightened risk perception. Overconfidence is particularly concerning in scenarios involving invasive therapies or high-stakes decisions, where inflated risk estimates may shift shared decision-making toward more aggressive management than warranted by true event probabilities.

Underappreciating calibration for threshold-based decisions. Calibration assumes greater clinical relevance than discrimination when models guide treatment thresholds—for instance, initiating statin therapy, anticoagulation, or intensive monitoring when predicted risk exceeds 20%. In these contexts, accurate absolute probabilities are essential for aligning recommendations with actual event likelihoods, enabling informed shared decision-making and appropriate risk-benefit discussions with patients. Even a model with excellent discrimination (high AUROC) can be misleading if poorly calibrated, as the predicted probability at the decision threshold may not correspond to the true risk, potentially leading to overtreatment or undertreatment despite strong ranking ability.

Misinterpreting binning artifacts, Calibration curves may appear well-aligned due to coarse binning of predicted probabilities, which can mask miscalibration within narrower probability ranges. This artifact arises because averaging over broad groups reduces sensitivity to local deviations from ideal calibration. Visual inspection of calibration plots should therefore be supplemented with formal statistical assessments, such as the Hosmer-Lemeshow test (though it has recognized limitations, including low power and sensitivity to binning choices) or more robust alternatives like expected calibration error (ECE) or Brier score decomposition. These supplementary measures provide a more reliable indication of calibration performance, particularly in regions of the probability spectrum most relevant to clinical thresholds.

While discrimination remains a valuable metric for assessing a model's ability to stratify risk, calibration directly influences the trustworthiness of absolute risk estimates used in everyday clinical decisions. Physicians should prioritize thorough evaluation of both properties, with particular attention to calibration when models support threshold-based interventions or direct patient communication of risk.